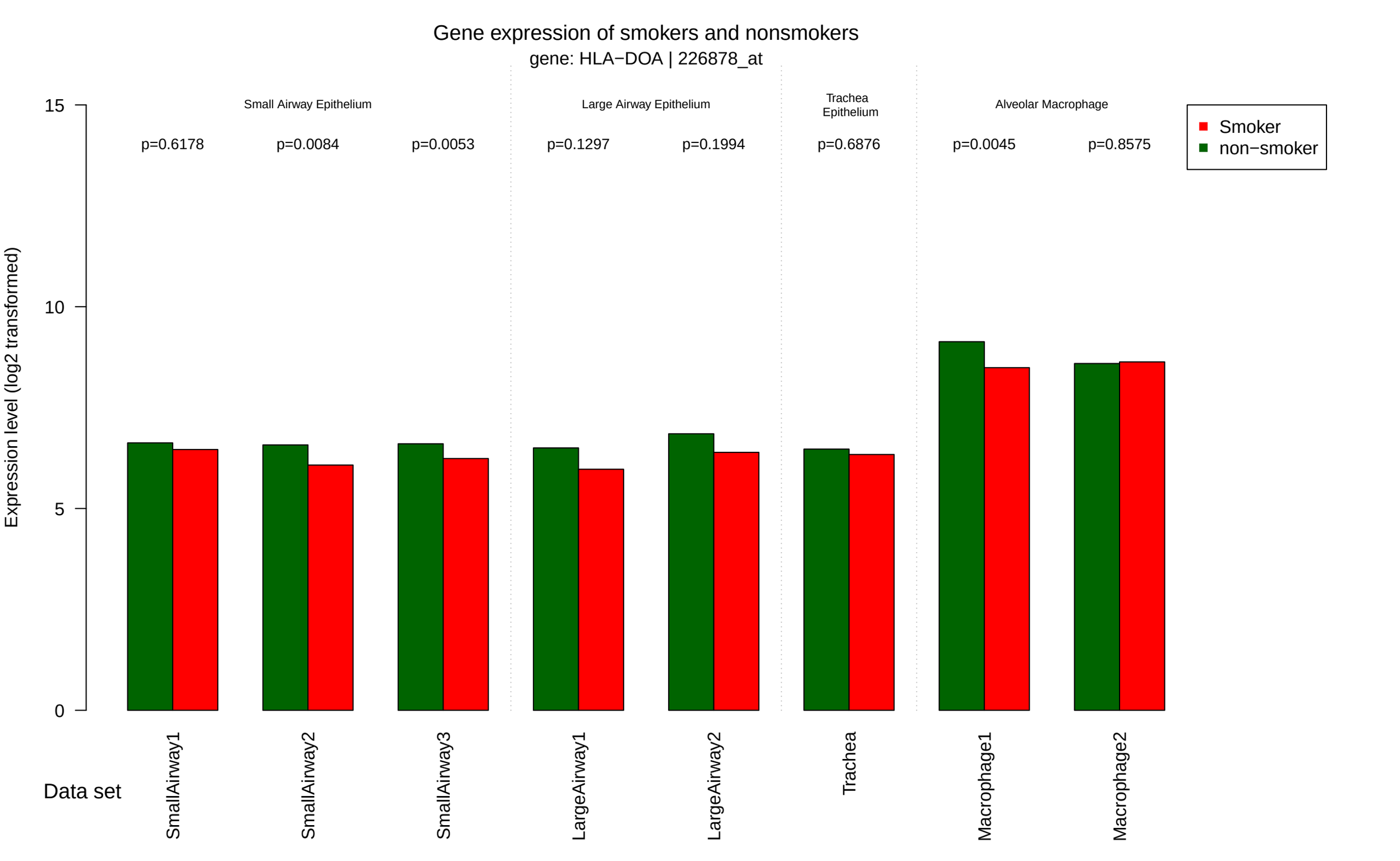
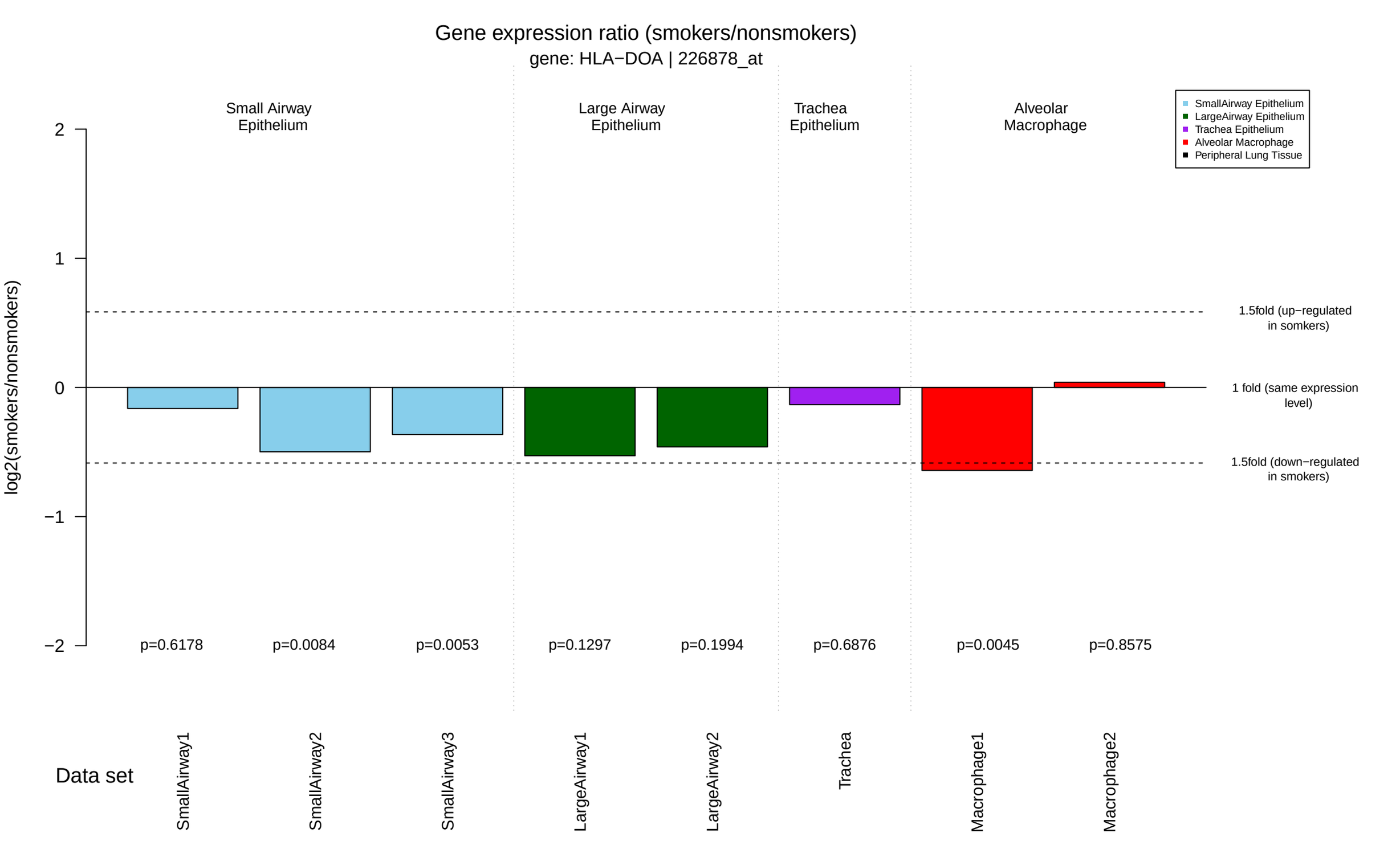
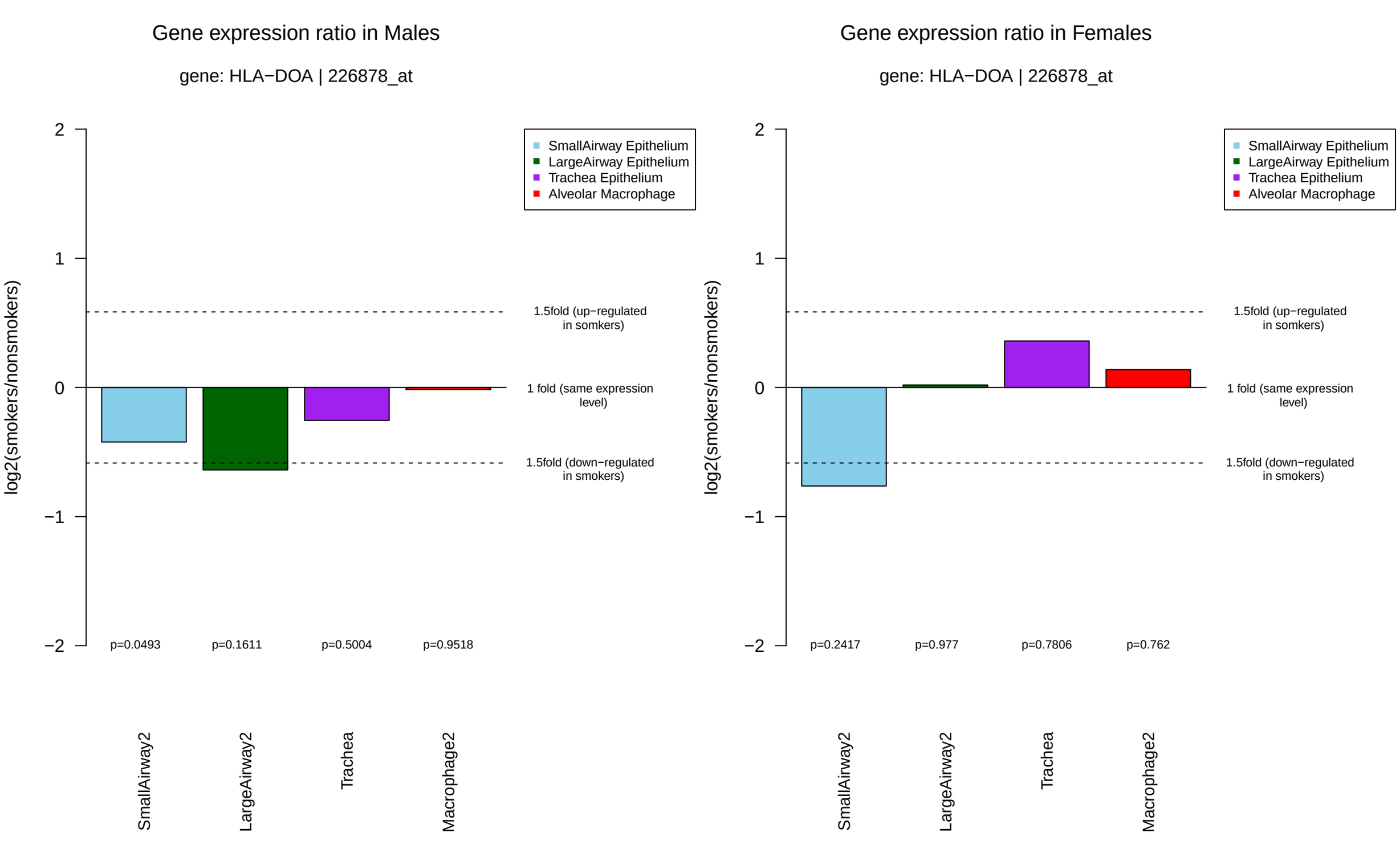
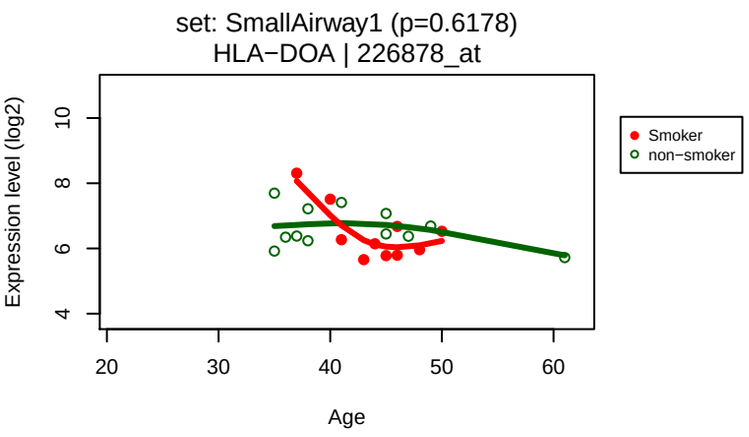
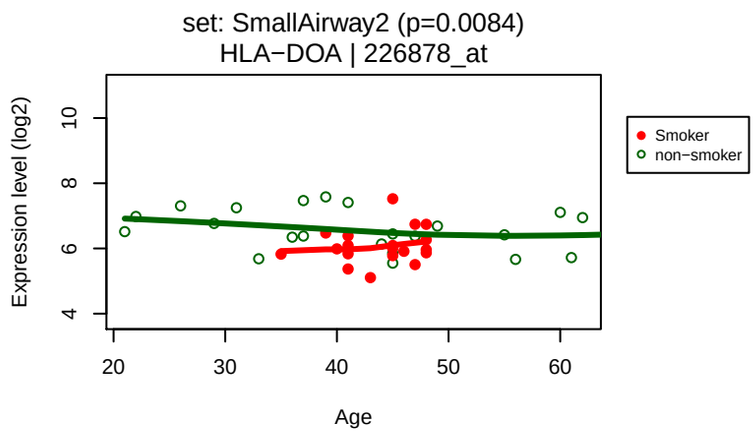
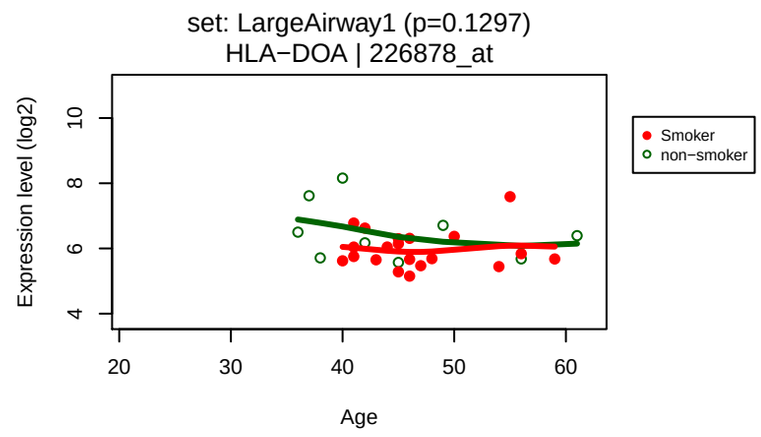

Gene expression of smokers and nonsmokers
gene: HLA−DOA | 226878_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6178
p=0.0084
p=0.0053
p=0.1297
p=0.1994
p=0.6876
p=0.0045
p=0.8575
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

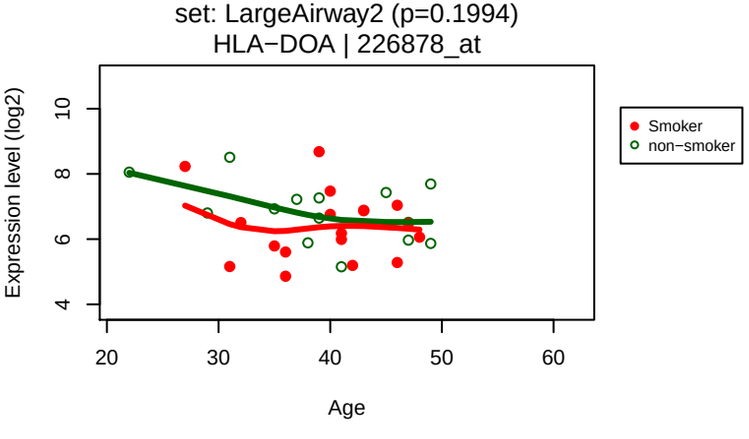

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DOA | 226878_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6178
p=0.0084
p=0.0053
p=0.1297
p=0.1994
p=0.6876
p=0.0045
p=0.8575
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

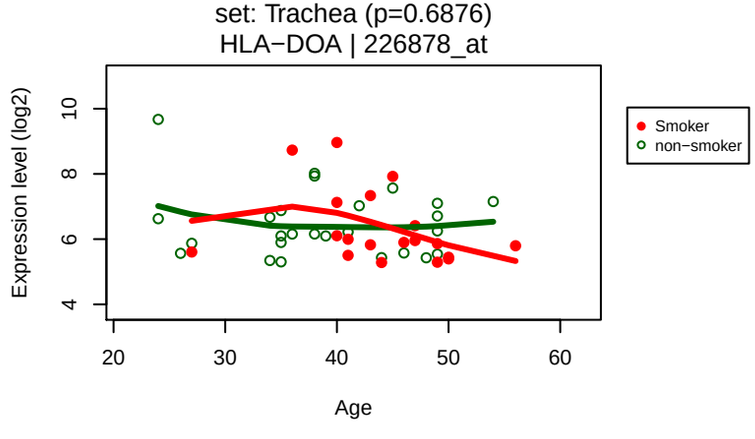

Gene expression ratio in Males
gene: HLA−DOA | 226878_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0493
p=0.1611
p=0.5004
p=0.9518
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DOA | 226878_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2417
p=0.977
p=0.7806
p=0.762
SmallAirway2
LargeAirway2
Trachea
Macrophage2

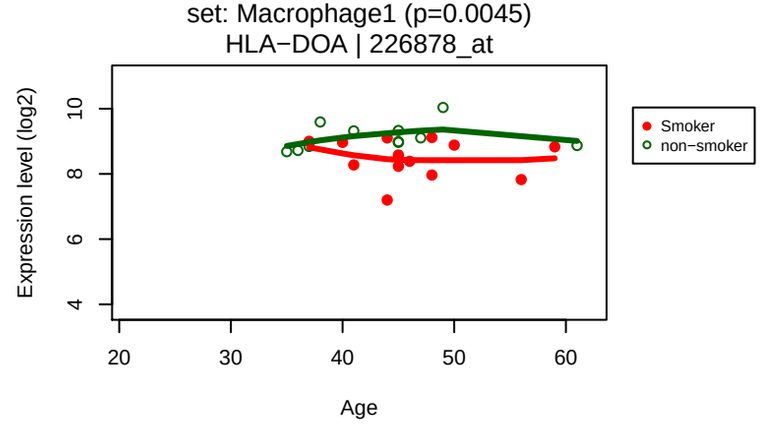

set: SmallAirway1 (p=0.6178)
HLA−DOA | 226878_at
Expression level (log2)
Age
Smoker
non−smoker

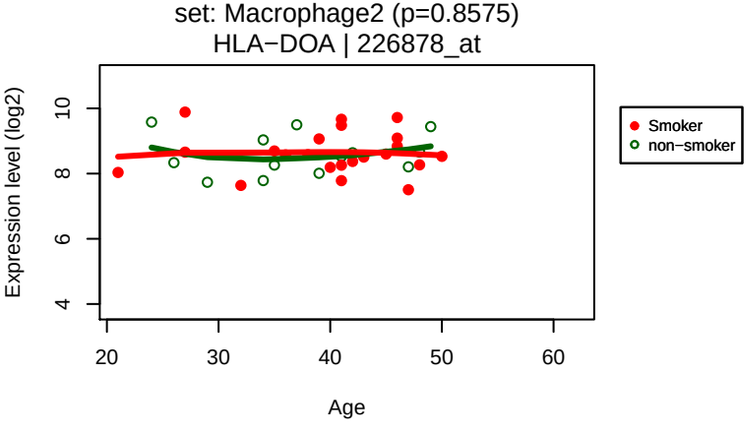

set: SmallAirway2 (p=0.0084)
HLA−DOA | 226878_at
Expression level (log2)
Age
Smoker
non−smoker

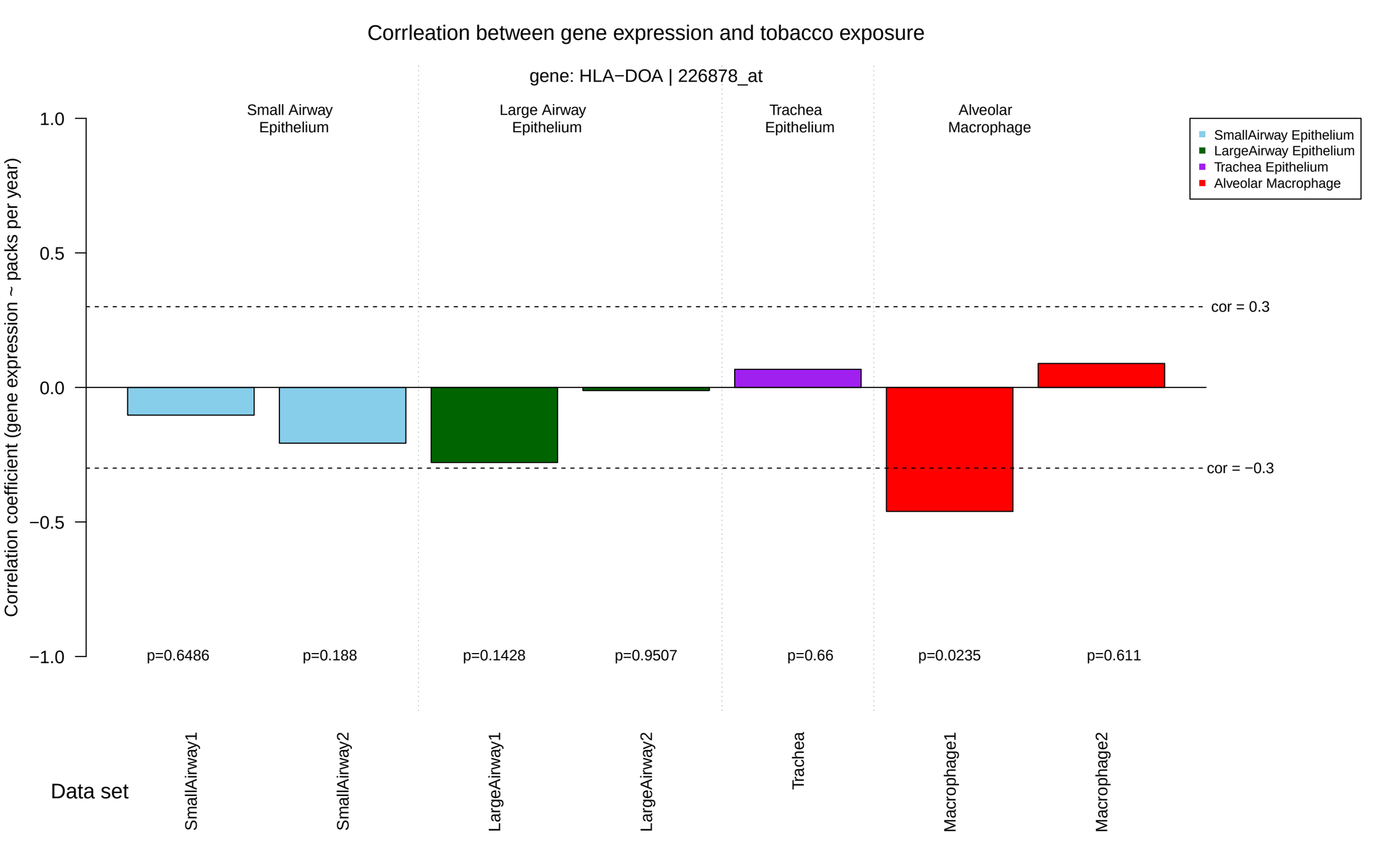

set: LargeAirway1 (p=0.1297)
HLA−DOA | 226878_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1994)
HLA−DOA | 226878_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6876)
HLA−DOA | 226878_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0045)
HLA−DOA | 226878_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8575)
HLA−DOA | 226878_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DOA | 226878_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6486
p=0.188
p=0.1428
p=0.9507
p=0.66
p=0.0235
p=0.611
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2